

Basic Principles of Medicine 2
Nervous System and Behavioral Sciences
Lecture No: 65

Flipped Classroom:
Sensory-Motor Integration

Sensory-Motor Integration

The slides and questions in this flipped classroom focus on providing an overview of select integrative and complex concepts that students often find challenging, and on developing higher-order thought processes required to effectively apply related knowledge .

Pre-Session Requirements

1. Review slides
2. Review and attempt to answer the MCQs provided

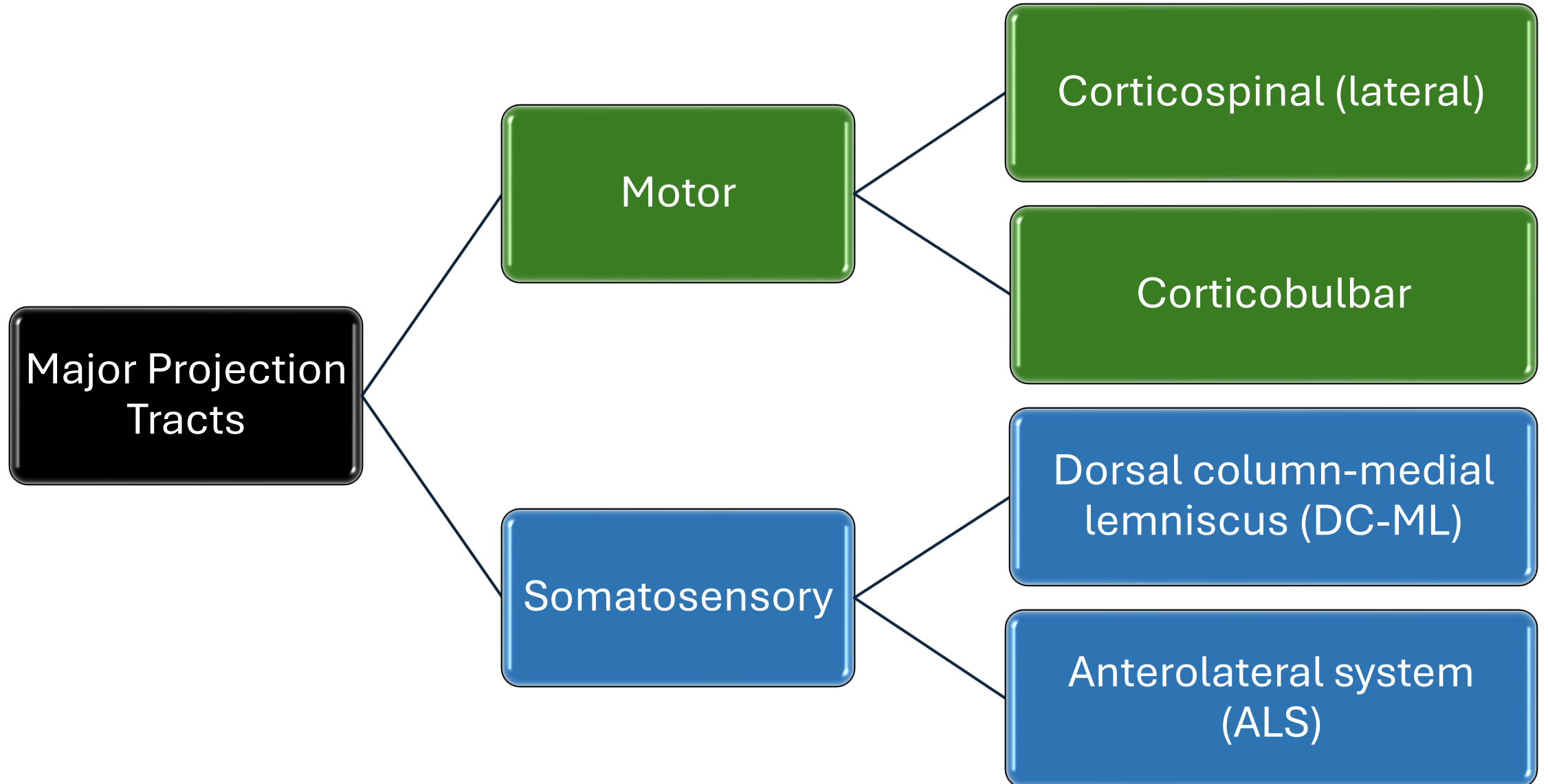
Learning Objectives

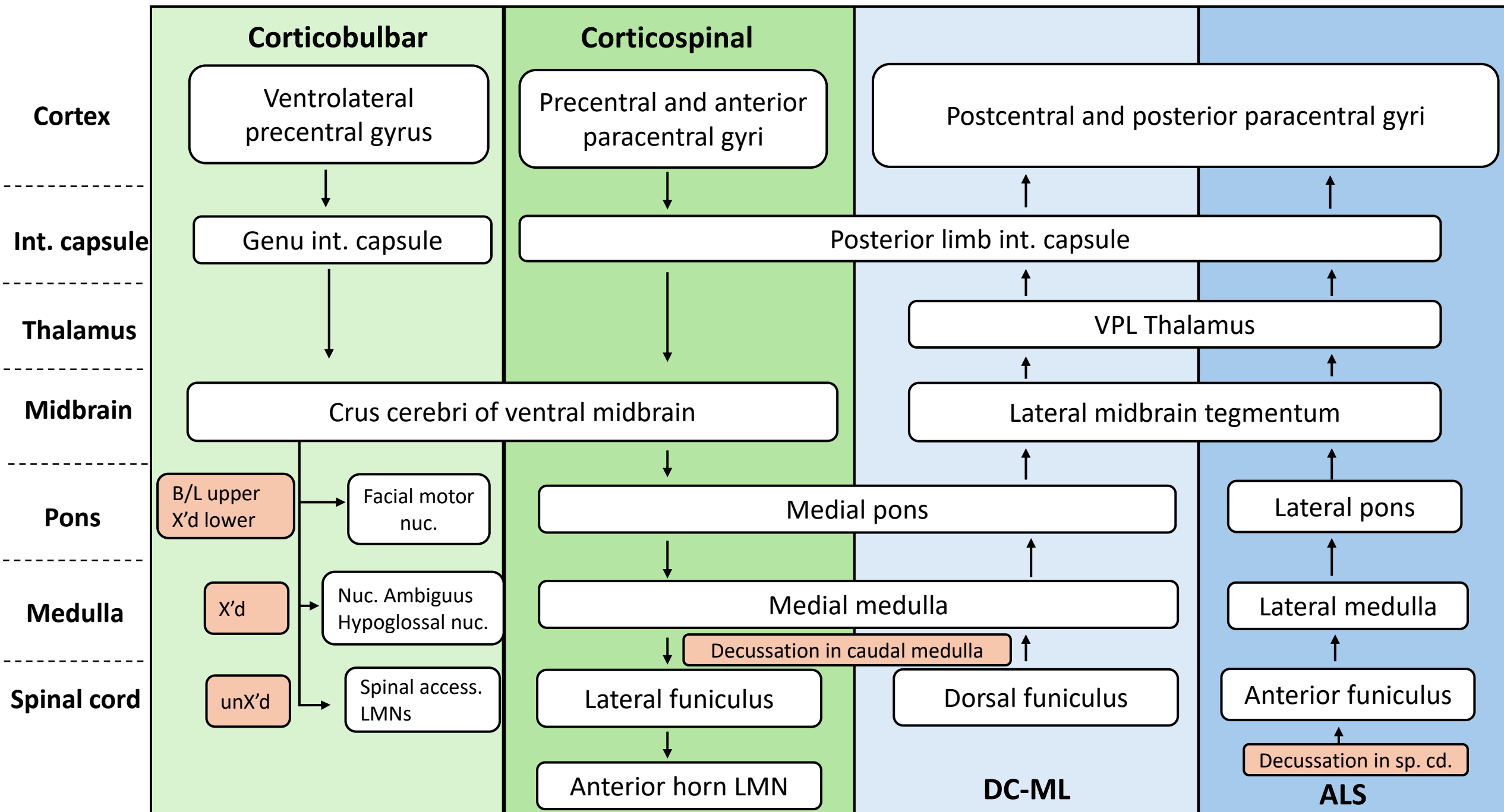
- SOM.MK.I.BPM2.4.NB.1.NSCI.0581 Integrate neuroanatomical knowledge of the corticospinal, corticobulbar, dorsal column-medial lemniscus and anterolateral system projection tracts
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- SOM.MK.I.BPM2.4.NB.1.NSCI.0584 Categorize saccadic, vestibulo-ocular and optokinetic eye movements as voluntary or reflex, describe their neuroanatomical pathways and describe their assessment in the neurological examination
- SOM.MK.I.BPM2.4.NB.1.NSCI.0585 Explain the phases and testing of physiological nystagmus
- SOM.MK.I.BPM2.4.NB.1.NSCI.0586 List four causes of pathological nystagmus and the describe expected related observations during neurological examination

Learning Objectives

- SOM.MK.I.BPM2.4.NB.1.NSCI.0587 Apply knowledge of the corticospinal, corticobulbar, dorsal column-medial lemniscus and anterolateral system tracts to localize a lesion in the central nervous system
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- SOM.MK.I.BPM2.4.NB.1.NSCI.0589 Apply knowledge of saccadic, vestibulo-ocular and optokinetic eye movements, and physiological and pathological nystagmus to localize a lesion in the central nervous system

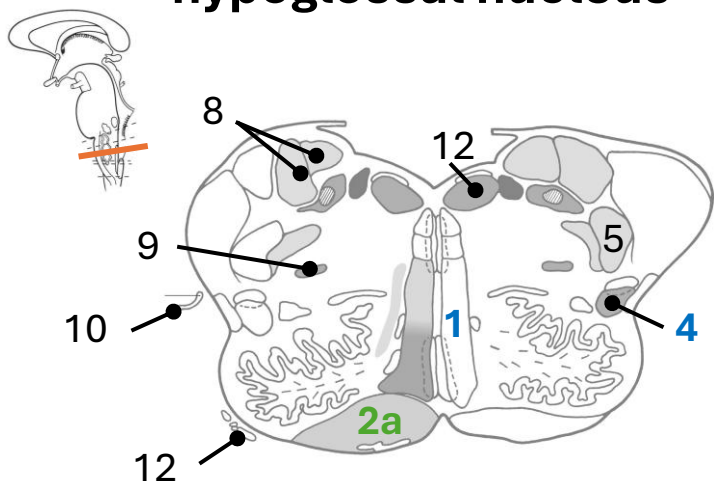
Projection Tracts Overview



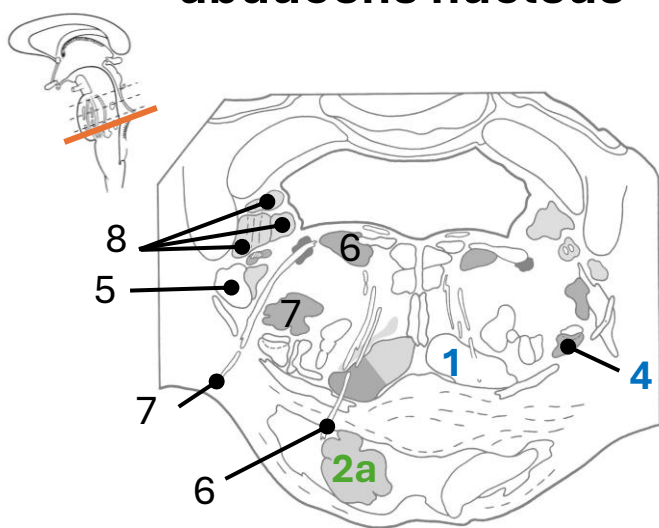


Representative Positioning Of Tracts In Brainstem

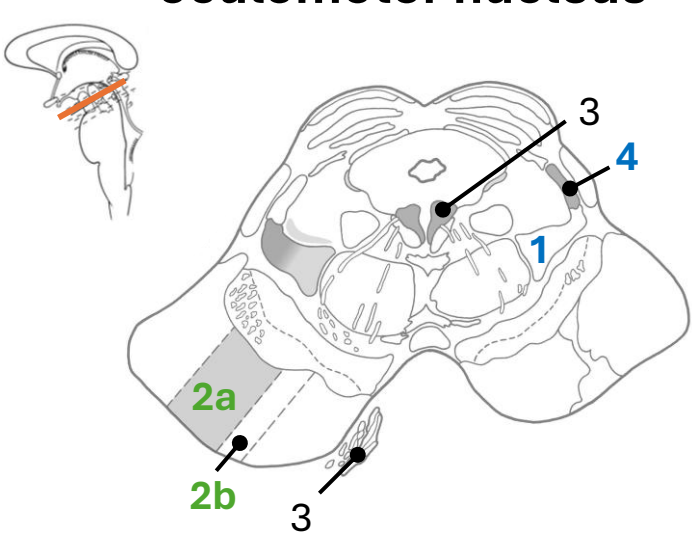
Medulla at level of hypoglossal nucleus



Pons at level of abducens nucleus



Midbrain at level of oculomotor nucleus



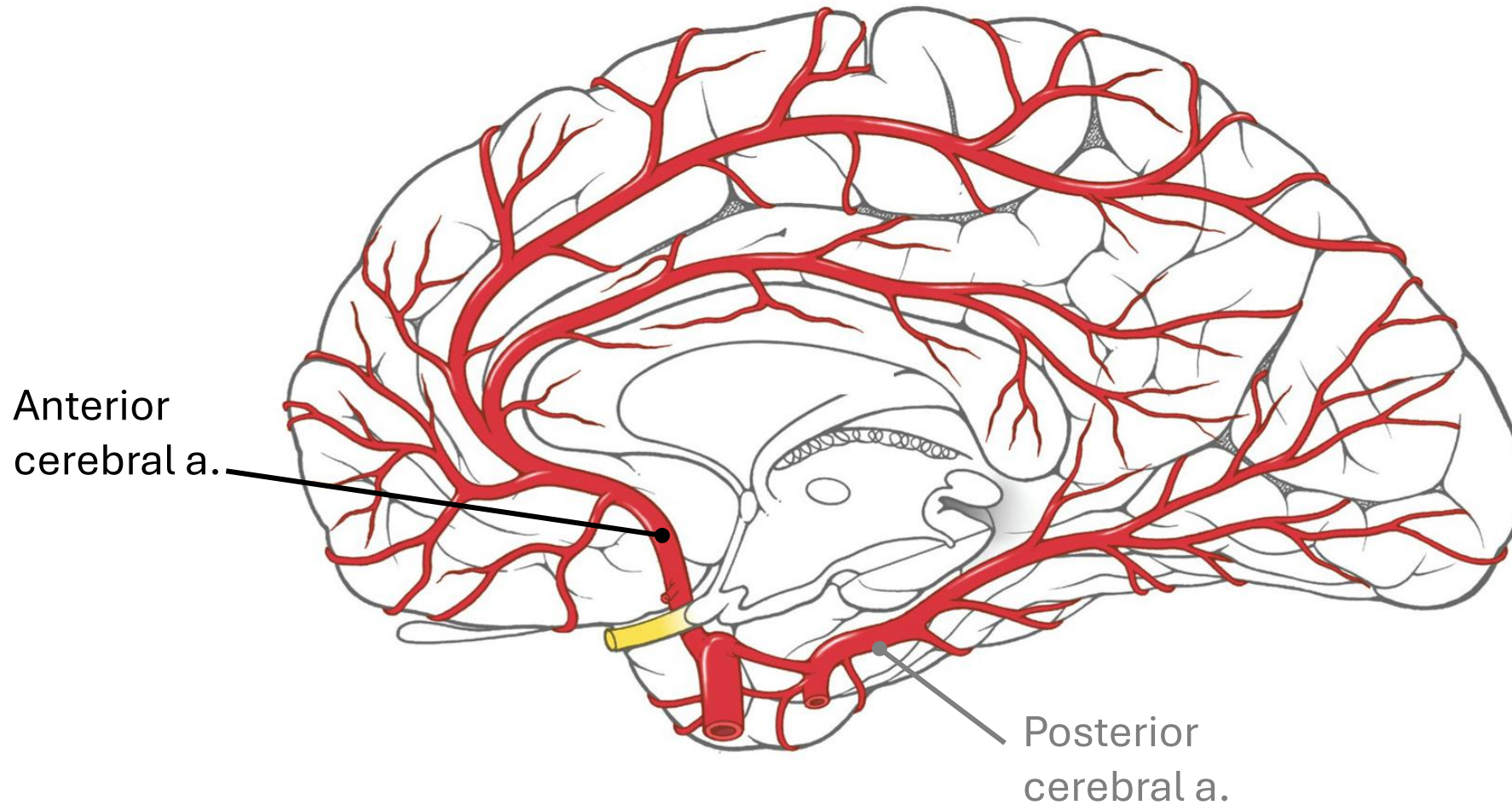
1	Medial lemniscus	5	Spinal trigeminal tract	9	Nucleus ambiguus
2a, 2b	Corticospinal (a), corticobulbar (b)	6	Abducens nucleus / nerve	10	Vagus nerve
3	Oculomotor nucleus / nerve	7	Facial motor nucleus / facial nerve	12	Hypoglossal nucleus / nerve
4	ALS	8	Vestibular nuclei		

Remember that all structures shown are normally present bilaterally!

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Anterior Cerebral Artery (ACA)



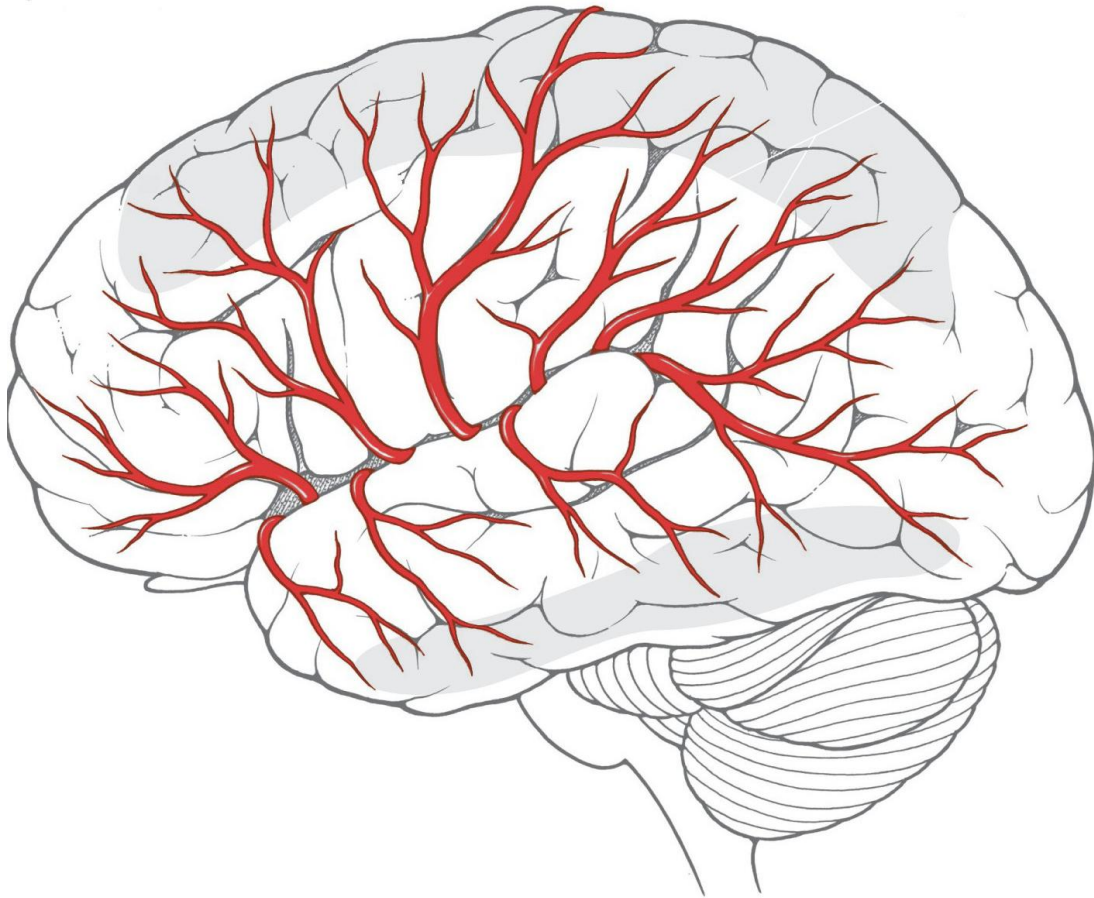
Medial surface

Review!

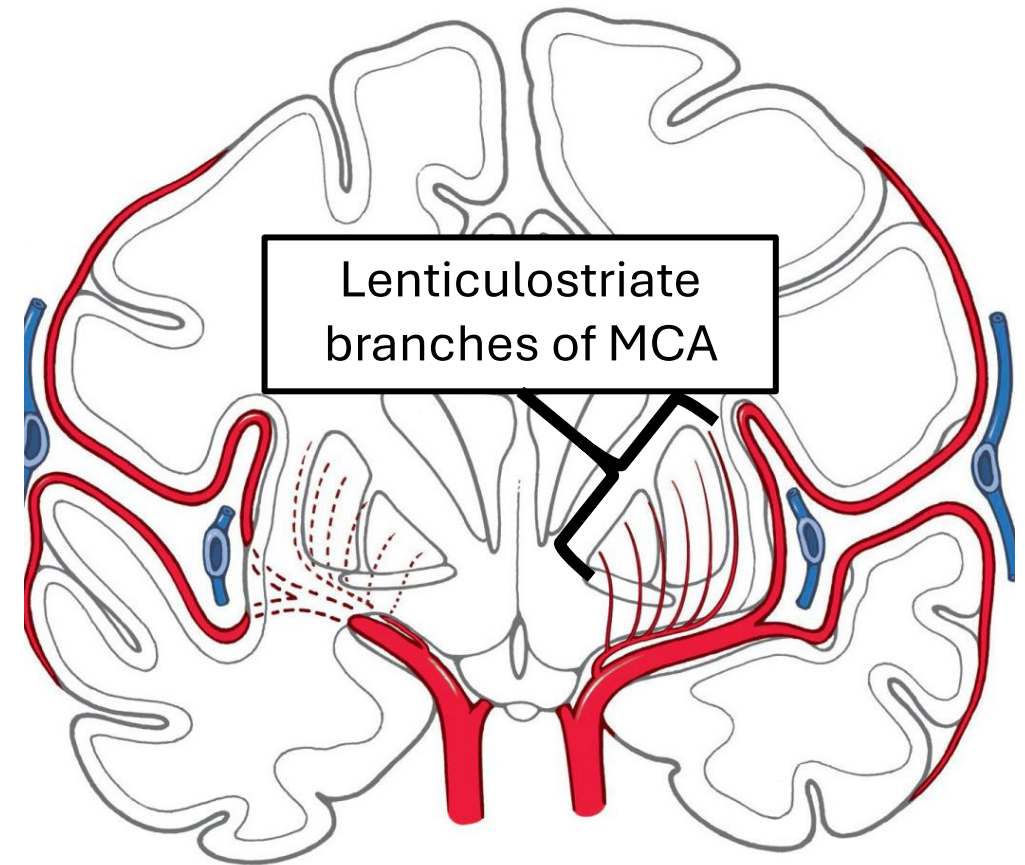
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Middle Cerebral Artery (MCA)

Cortical branches of MCA

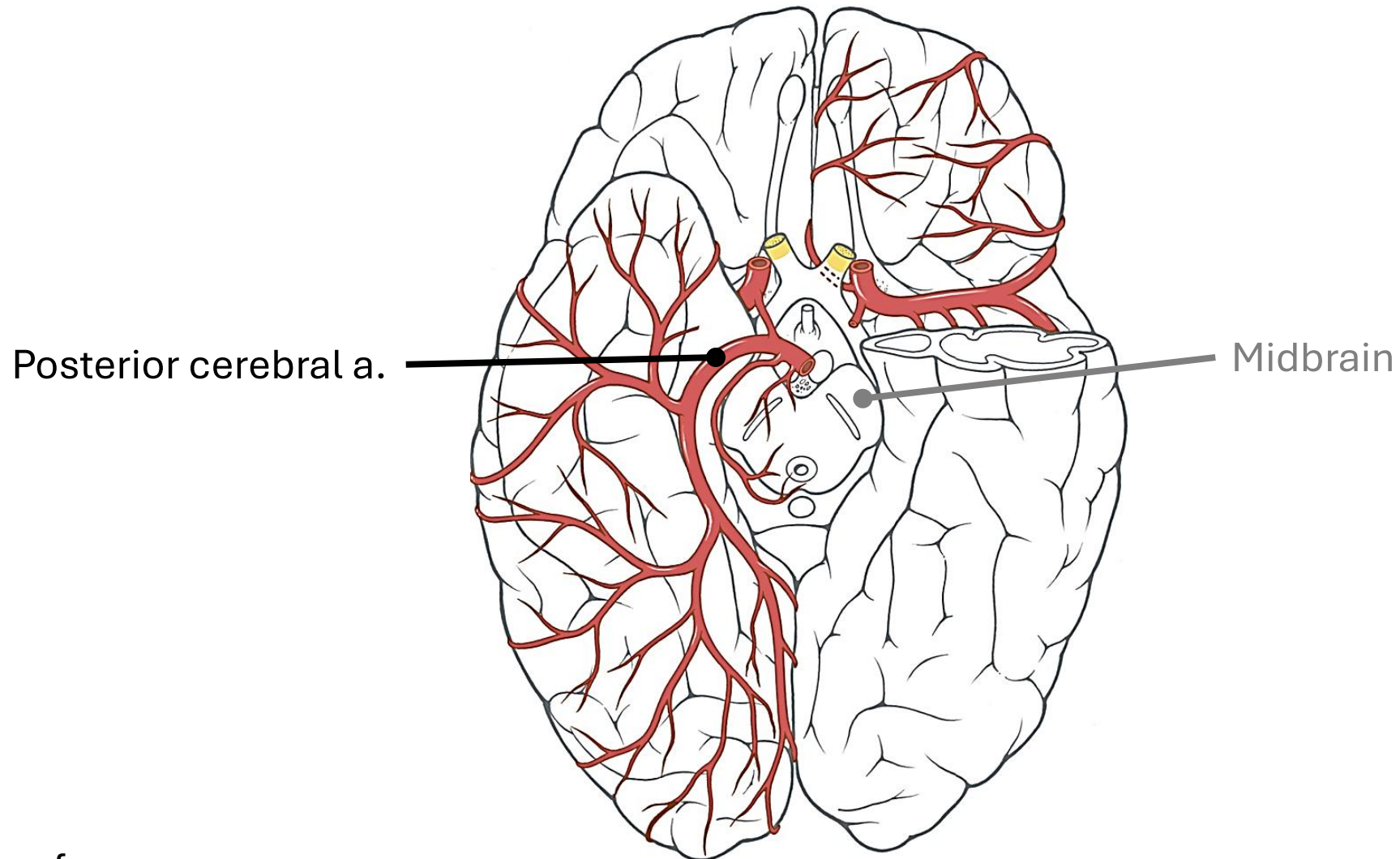


Lateral surface



Review!

Posterior Cerebral Artery (PCA)



Inferior surface

Review!

NSCI.0072

Cerebral Artery Syndromes

Artery Affected	Possible Features
Anterior	Contralateral lower limb motor and somatosensory deficits, urinary incontinence (automatic bladder).
Middle - cortical branches	Contralateral face and upper limb motor and somatosensory deficits. Aphasia if dominant hemisphere affected*. Gaze deviation / paralysis if frontal eye field (FEF) affected. *Note: the aphasias will be covered in detail in NB Lecture 73
Middle - lenticulostriate	Contralateral motor and somatosensory deficits of face, upper limb, and lower limb.
Posterior	Contralateral homonymous hemianopia with macular sparing. Alexia without agraphia (dominant hemisphere) / prosopagnosia (non-dominant hemisphere). Contralateral motor deficits, ipsilateral CN III involvement if midbrain branches affected.

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Lateral Pontine vs Lateral Medullary Syndrome:

Similarities

Structure Damaged	Deficit
Anterolateral system	Contralateral loss / decrease of pain and temperature sense for the body
Spinal trigeminal tract and nucleus	Ipsilateral loss / decrease of pain and temperature sense for the face
Hypothalamospinal fibers	Ipsilateral Horner syndrome
Cerebellar peduncles (superior and middle in pons, inferior in medulla)	Ataxia, unsteady gait, falls towards side of lesion
Vestibular nuclei	Nausea, nystagmus, vertigo

Lateral Pontine vs Lateral Medullary Syndrome:

Differences

Lateral Pontine Syndrome

Structure Damaged	Deficit
Trigeminal motor nucleus (mid level)	Ipsilateral paralysis of masticatory muscles
Facial motor nucleus (caudal)	Ipsilateral facial paralysis (LMN)
Cochlear nerve (caudal)	Ipsilateral sensorineural hearing loss

Lateral Medullary Syndrome

Structure Damaged	Deficit
Nucleus ambiguus, roots of CN IX, X	Dysphagia, hoarseness (dysarthria), soft palate paralysis, diminished gag reflex

Medial Pontine vs Medial Medullary Syndrome:

Similarities

Structure Damaged	Deficit
Corticospinal fibers	Contralateral motor deficits (upper motor neuron syndrome, UMNS)
Medial lemniscus	Contralateral loss / decrease of discriminative touch, vibration and proprioception for the body

Medial Pontine vs Medial Medullary Syndrome:

Differences

Medial Pontine Syndrome

Structure Damaged	Deficit
Abducens nerve fibers or nucleus (caudal)	Ipsilateral lateral rectus paralysis (medial deviation on primary gaze)

Medial Medullary Syndrome

Structure Damaged	Deficit
Hypoglossal nerve / nucleus	Ipsilateral deviation of the tongue when protruded (atrophy, fasciculations)

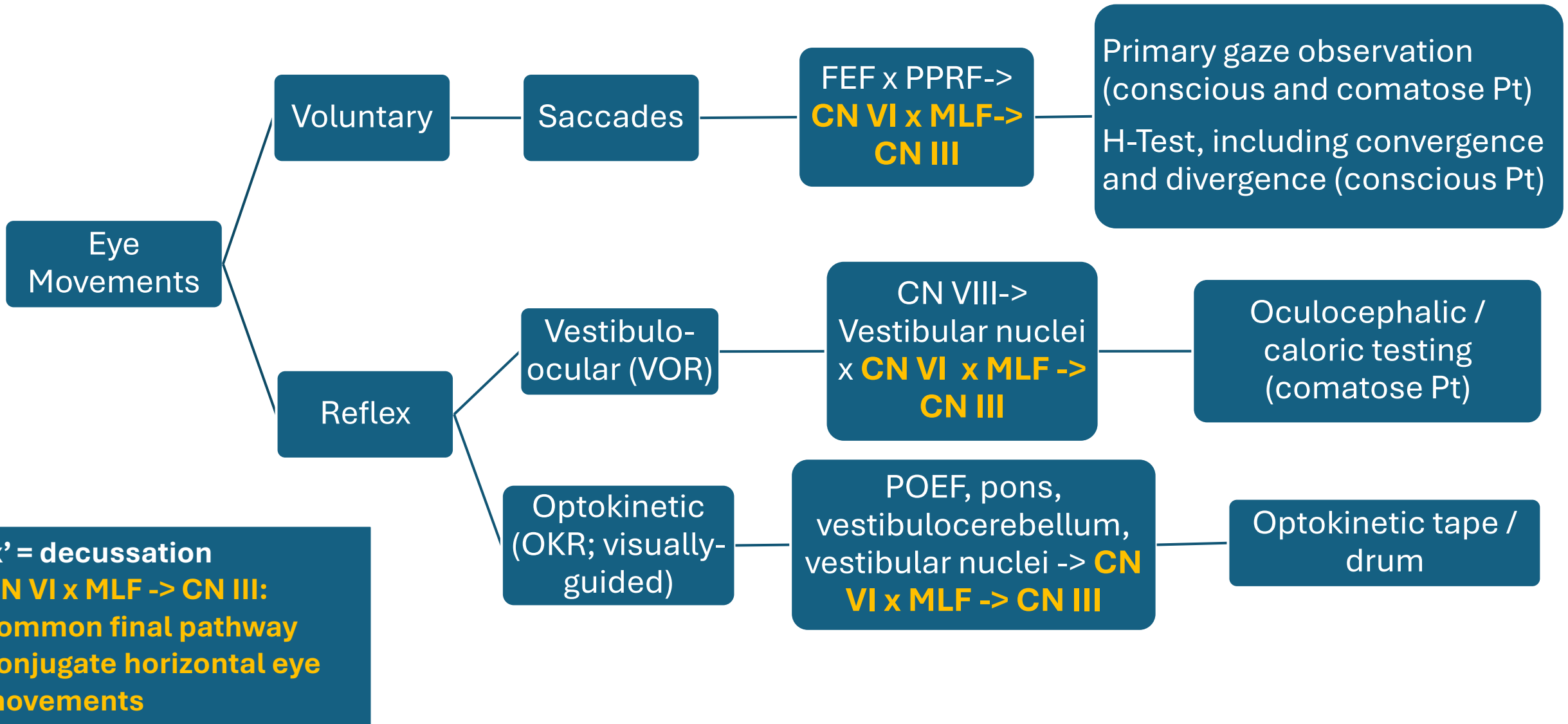
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Eye Movements: Terminology

- **Primary gaze:** Straight-ahead gaze
- **Gaze paralysis** to the left: Inability to look to the left
- **Gaze deviation** to the left: Eyes are tonically or spontaneously directed to a leftward gaze position

Eye Movements: Big Picture



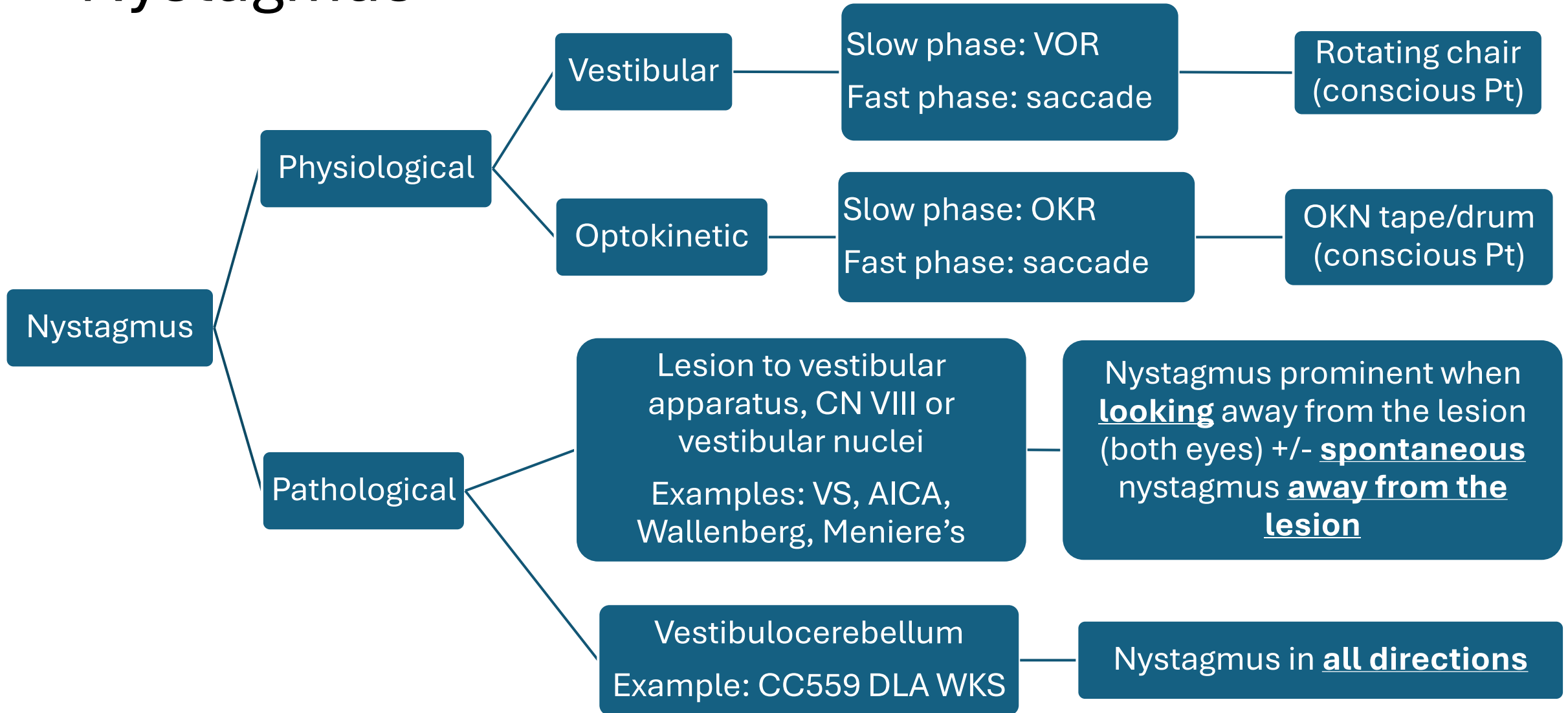
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Nystagmus: Terminology

- **Primary gaze:** straight-ahead gaze
- **Nystagmus:** rapid, alternating movement of the eyes. Two phases: first, slow, REFLEX phase and second, fast, SACCADIC, reset phase. Direction of nystagmus is determined by the direction of the fast / reset phase.
- **Left-beating = leftward = leftward moving = nystagmus to the left** (i.e., the fast, reset, saccadic phase is to the left). By definition, this means that the slow, reflex phase is to the right.

Nystagmus



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Clicker Question Next!

Please prepare to answer
the clicker question
on the next slide.

Apply Your Knowledge

- Prior to the Flipped Classroom session, attempt to answer the questions that follow.
- Questions will be presented and discussed in class, but with limited time to answer – please come with your answers prepared!

1. A 57-year-old man is brought to the emergency department by ambulance because of sudden onset numbness and weakness on one side of his body. Neurologic examination shows deficits in touch perception and reduced power on the right side of the body. No deficits in nociception are noted. Additionally, hyperreflexia is evident in the upper and lower extremities on the right. Branch occlusion of which of the following arteries most likely gives rise to the deficits seen in this patient?

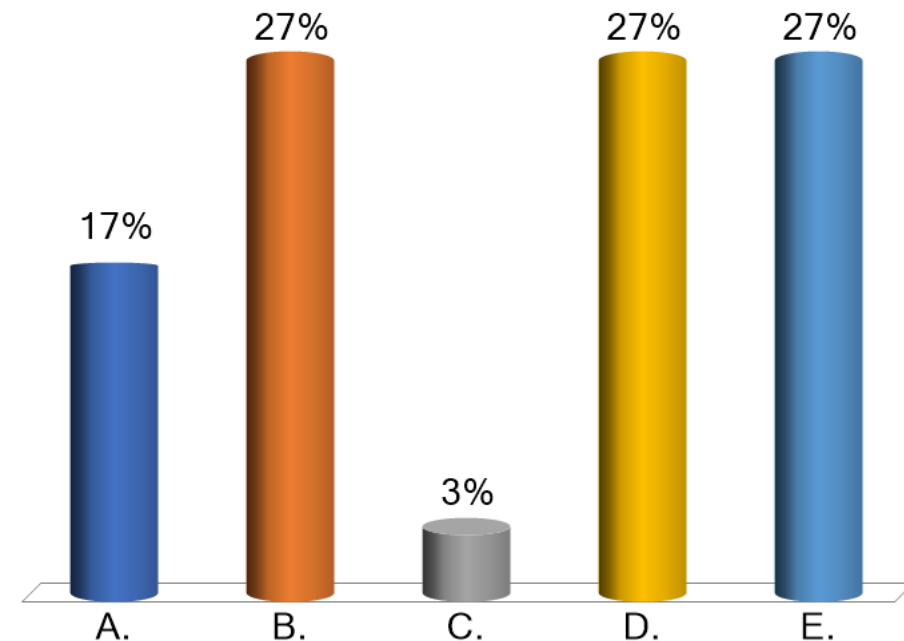
A. Anterior cerebral

😊 B. Anterior spinal

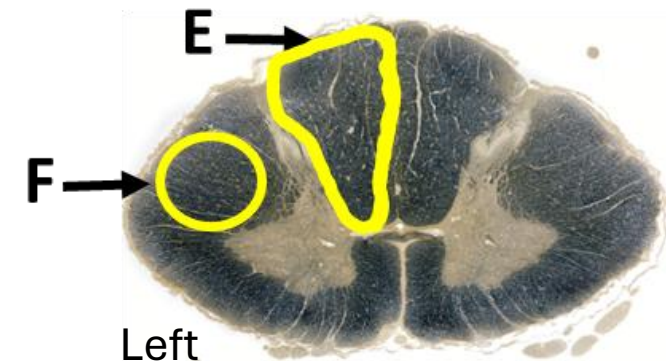
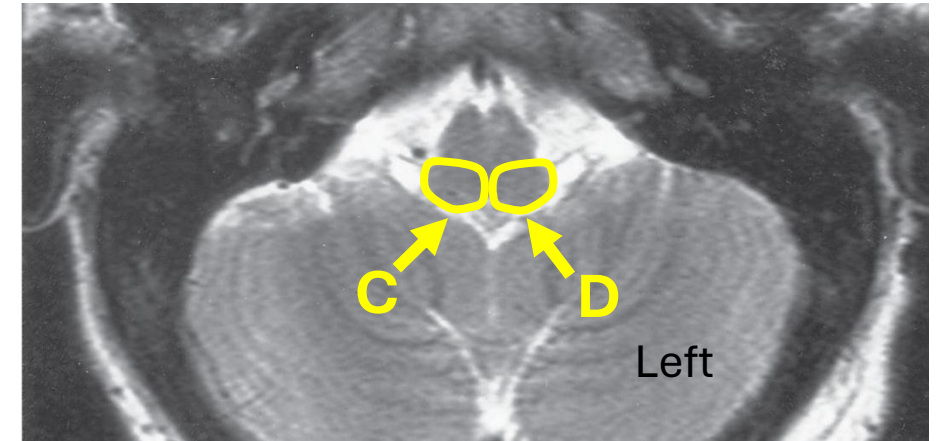
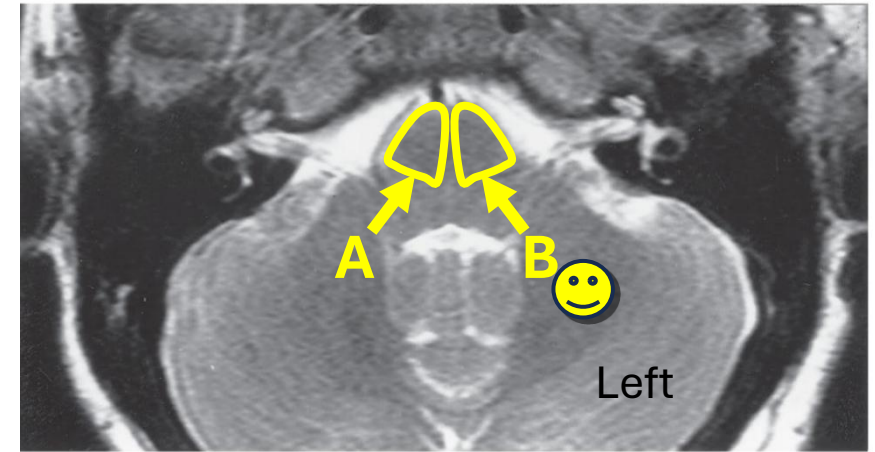
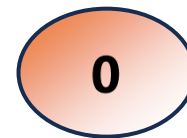
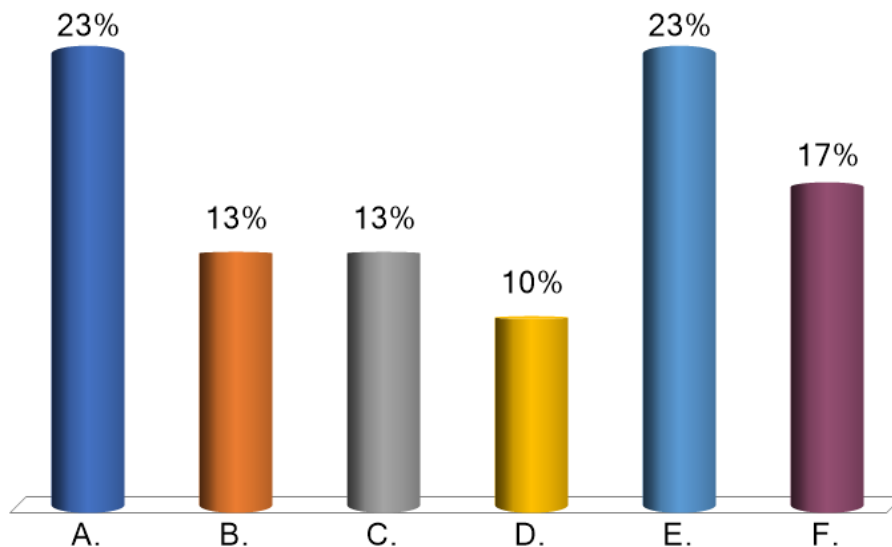
C. Middle cerebral

D. Posterior inferior cerebellar

E. Posterior spinal

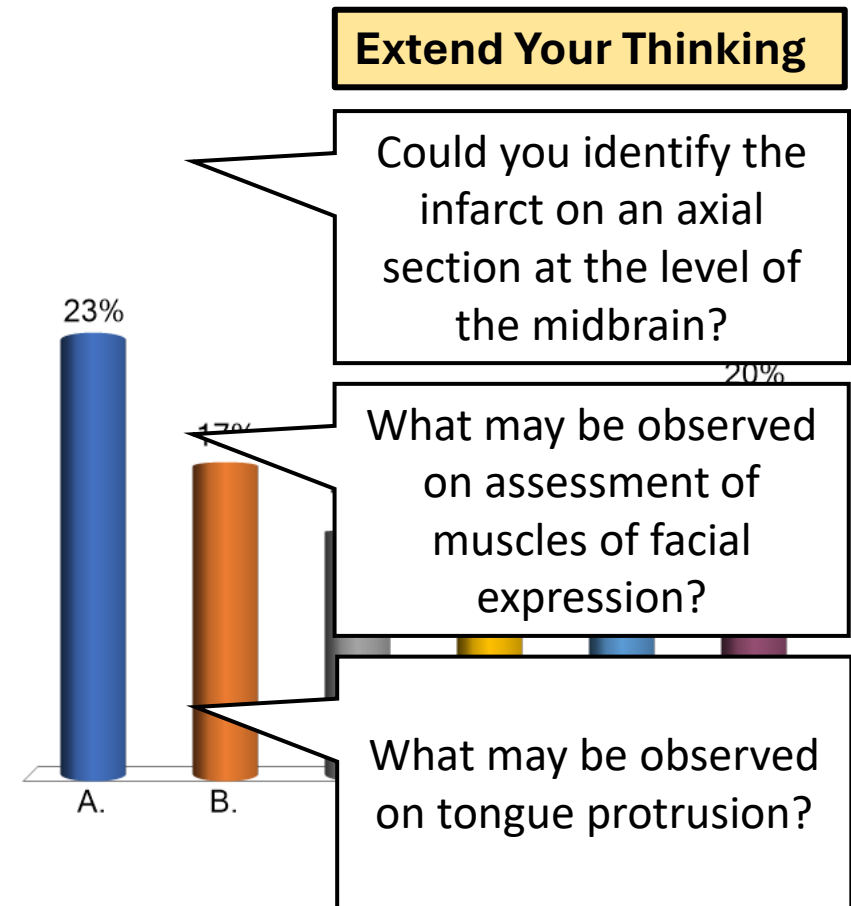


2. A 57-year-old man is brought to the emergency department because of sudden onset numbness and weakness on one side of his body. Neurologic examination shows deficits in touch perception and reduced power on the right side of the body. No deficits in nociception are noted. Additionally, hyperreflexia is evident in the upper and lower extremities on the right. Infarction affecting which of the areas indicated below in healthy individuals most likely gives rise to the deficits seen in this patient?



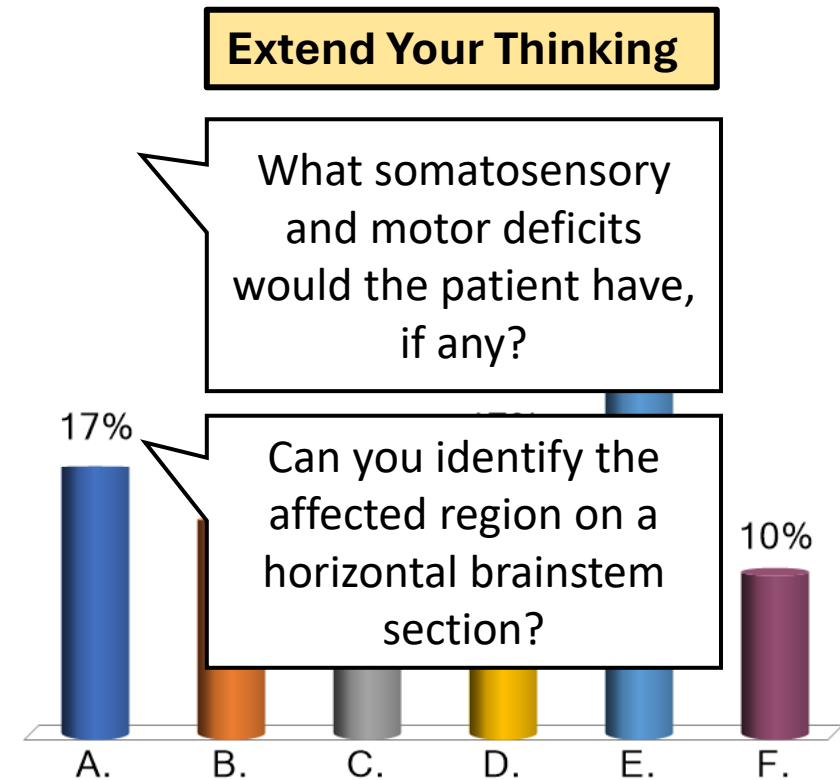
3. A 53-year-old woman is brought to the emergency department because of sudden onset weakness on the left side of her body. Neurologic examination shows reduced power and increased tone in the left hemibody and an unreactive, dilated pupil on the right. Neuroimaging shows infarction in the territory of the right posterior cerebral artery. Which of the following is most likely observed in the patient?

- A. Left tongue atrophy
- B. Right tongue atrophy
- C. Broca's aphasia
- D. Wernicke's aphasia
- 😊 E. Left homonymous hemianopia
- F. Right homonymous hemianopia



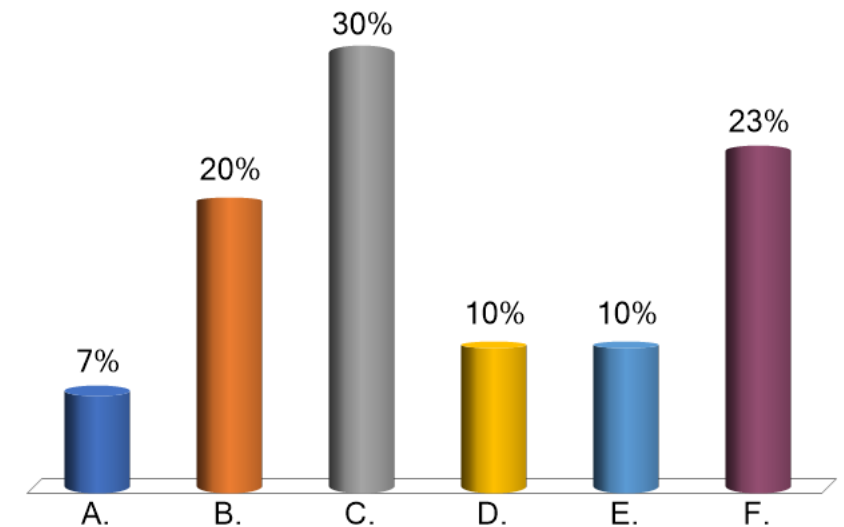
4. A 63-year-old woman comes to the physician because of problems hearing which began suddenly three days ago. Tuning fork tests show sensorineural hearing loss on the right. Weakness of the muscles of facial expression on the right is also noted. What is the most likely cause of the patient's symptoms, and the direction of pathological nystagmus that would be observed?

- A. Left AICA occlusion, nystagmus to the right
- 😊 B. Right AICA occlusion, nystagmus to the left
- C. Left vestibular schwannoma, nystagmus to the right
- D. Right vestibular schwannoma, nystagmus to the left
- E. Left PICA occlusion, nystagmus to the right
- F. Right PICA occlusion, nystagmus to the left



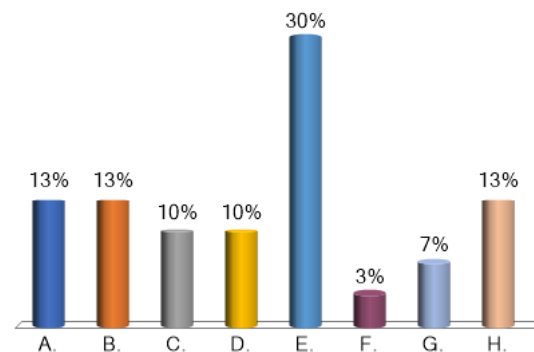
5. A 68-year-old man is brought to the emergency department by his wife because of sudden onset weakness. Neurologic examination shows reduced power and hypertonia of the right upper extremity. Language is telegraphic in nature, but the patient can follow commands and is fully oriented. Repetition of words and phrases is impaired. Which of the following additional deficits is most likely observed?

- A. Loss of somatosensation in right upper extremity
- B. Loss of somatosensation in left upper extremity
- C. Superior right quadrantanopia
- D. Superior left quadrantanopia
- 😊 E. Weakness of the right lower face
- F. Weakness of the left lower face



6. A 55-year-old unconscious man is brought to the emergency department by ambulance following a motor vehicle collision. Caloric testing is performed by injecting warm water first into the man's right external auditory meatus and then into the left. The findings are depicted in the image below. What is the most likely location of the lesion?

- A. Left vestibular nucleus
- B. Left MLF
- C. Left abducens nerve
- D. Left oculomotor nerve
- E. Right vestibular nucleus
- F. Right MLF
- 😊 G. Right abducens nerve
- H. Right oculomotor nerve



Warm water in right ear

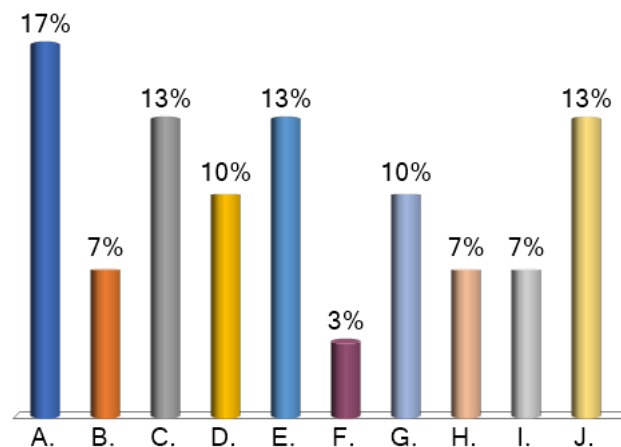
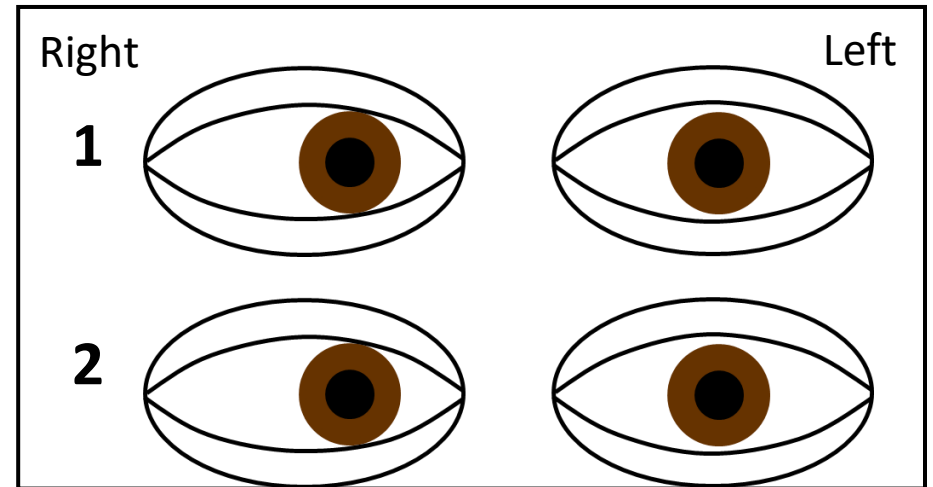


Warm water in left ear



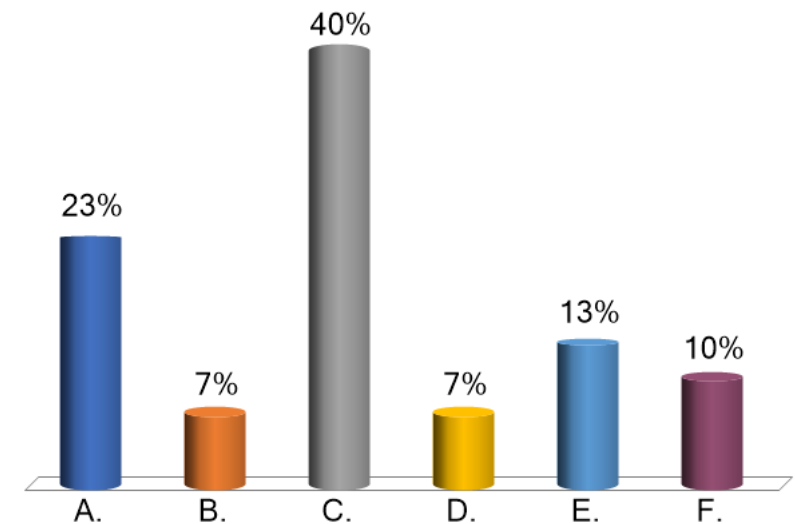
7. A 77-year-old man is brought to the emergency department by ambulance because of loss of consciousness 2 hours ago. Pupillary light reflex is bilaterally normal. The image below shows (1) the patient's primary gaze and (2) extraocular response when cold water is injected into the right ear during caloric testing. What is the most likely location of the lesion?

- A. Left oculomotor nerve
- B. Left abducens nucleus
- C. Left abducens nerve
- D. Left MLF
- E. Left vestibular nuclei
- F. Right oculomotor nerve
- 😊 G. Right abducens nucleus
- H. Right abducens nerve
- I. Right MLF
- J. Right vestibular nuclei



8. A 27-year-old man is brought to the emergency department by his father having fallen down a flight of stairs. The man briefly lost consciousness but is alert and oriented on arrival. Trauma to the right side of the head is evident. Before neurological examination can be carried out, the man loses consciousness again and exhibits left decerebrate posture. Plantar response is extensor on the left and flexor on the right. Corneal reflex is intact bilaterally. Neuroimaging shows a frontotemporal epidural hemorrhage and uncal herniation. Which of the following combinations of findings will most likely be seen on further examination of the patient?

- A. Eyes deviate to the right, tongue deviates left on protrusion
- B. Eyes deviate to the left, tongue deviates right on protrusion
- C. Eyes deviate to the right, right knee jerk hyperreflexia
- D. Eyes deviate to the left, left knee jerk hyperreflexia
- 😊 E. Eyes deviate to the right, right pupil dilated and unreactive
- F. Eyes deviate to the left, left pupil dilated and unreactive



End